
GENERAL RELATIVITY

Tutorial problem set 16, 17.01.2020.

■ **PROBLEM 39** The ergosphere and ergoregion of the Kerr black hole.

- (a) Determine the boundary of the region in Kerr space-time inside of which it is impossible for an observer to remain at fixed coordinates (r, θ, ϕ) . Express this boundary in a form $r(\theta)$. Note that this region extends outside of the event horizon at $r_+ = MG_N + \sqrt{M^2 G_N^2 - a^2}$. The region of spacetime outside of the horizon where there are no stationary observers possible is called the ergoregion and its boundary is called the ergosphere. It is an effect of the rotation of the black hole. Sketch the ergoregion, ergosphere and the horizon.
- (b) Although static observers are not possible inside the ergosphere, observers with fixed just r and θ coordinates that rotate with constant angular velocity as seen from infinity ($d\phi/dt = \Omega = \text{const.}$) along with the black hole are allowed. Show that it is so.
- (c) Show that the minimal angular velocity of the massless particle at the horizon r_+ is

$$\Omega_H = \left(\frac{d\phi}{dt} \right)_{\min} (r_+) = \frac{a}{r_+^2 + a^2} . \quad (39.1)$$

- (d) Show that the Killing vector $\chi^\mu = K^\mu + \Omega_H R^\mu$ (where K^μ and R^μ are defined in problem 1) has a Killing horizon, and determine where it is.

■ **PROBLEM 40** Gauge invariant perturbations.

Consider a slightly perturbed Minkowski space,

$$g_{\mu\nu}(x) = \eta_{\mu\nu} + h_{\mu\nu}(x) , \quad (40.1)$$

where $\eta_{\mu\nu} = \text{diag}(-1, 1, 1, 1)$ is the canonical Minkowski metric, and $h_{\mu\nu}$ is the small perturbation. The perturbation can be decomposed into parts that transform differently under *spatial rotations*, which is given in the usual way as

$$t \rightarrow \tilde{t} = t , \quad x^i \rightarrow \tilde{x}^i = R^i_j x^j , \quad \mathbf{R}\mathbf{R}^\top = \mathbf{1}, \quad \det(\mathbf{R}) = 1 . \quad (40.2)$$

- (a) Show that under spatial rotations: (i) h_{00} transforms as a scalar, (ii) h_{0i} transforms as a vector, (iii) h_{ij} transforms as a tensor.

According to the Helmholtz's theorem on vector calculus, h_{0i} can be decomposed into two distinct parts, a transverse (divergence-free) part, and a longitudinal (curl-free) part,

$$h_{0i} = B_i + \partial_i S , \quad (40.3)$$

where $\partial^i B_i = 0$ and $\nabla \times (\nabla S) = 0$, and where we have used the result that the curl-free part can be written as a gradient of some scalar function S . In a similar way, a 2-tensor under spatial rotations can be decomposed as

$$h_{ij} = 2\delta_{ij}\psi + 2\partial_i\partial_j E + \partial_i F_j + \partial_j F_i + h_{ij}^{TT} . \quad (40.4)$$

Here ψ and E are scalar functions, F_j is a transverse (divergence-free) vector, and h_{ij}^T is a *transverse traceless tensor*,

$$\partial^i h_{ij}^{TT} = 0 , \quad (h^{TT})^i_i = 0 . \quad (40.5)$$

It is also customary to relabel the scalar under spatial rotations,

$$h_{00} = 2\phi . \quad (40.6)$$

- (b) Convince yourself that this decomposition preserves the number of degrees of freedom originally contained in $h_{\mu\nu}$.
- (c) Under the infinitesimal diffeomorphism transformation (which is an active coordinate transformation, as opposed to a passive one), the metric perturbation transforms as

$$h_{\mu\nu}(x) \rightarrow \tilde{h}_{\mu\nu}(x) = h_{\mu\nu}(x) + \partial_\mu \xi_\nu(x) + \partial_\nu \xi_\mu(x) , \quad (40.7)$$

where $h_{\mu\nu}$ and ξ_μ are of the same order.

Utilizing the fact that ξ_μ can be decomposed with respect to spatial rotations into a scalar ξ_0 , and a vector $\xi_i = \xi_i^T + \partial_i \xi^S$, derive the following transformation rules for $h_{00}, B_i, S, \psi, E, F_i$ and h_{ij}^T under gauge (diffeomorphism) transformation (40.7)

$$\phi(x) \rightarrow \tilde{\phi}(x) = \phi(x) + \partial_0 \xi_0(x) , \quad (40.8)$$

$$S(x) \rightarrow \tilde{S}(x) = S(x) + \xi_0(x) + \partial_0 \xi^S(x) , \quad (40.9)$$

$$\psi(x) \rightarrow \tilde{\psi}(x) = \psi(x) , \quad (40.10)$$

$$E(x) \rightarrow \tilde{E}(x) = E(x) + \xi^S(x) , \quad (40.11)$$

$$B_i(x) \rightarrow \tilde{B}_i(x) = B_i(x) + \partial_0 \xi_i^T(x) , \quad (40.12)$$

$$F_i(x) \rightarrow \tilde{F}_i(x) = F_i(x) + \xi_i^T(x) , \quad (40.13)$$

$$h_{ij}^{TT}(x) \rightarrow \tilde{h}_{ij}^{TT}(x) = h_{ij}^{TT}(x) . \quad (40.14)$$

- (d) Show that, apart from h_{ij}^{TT} and ψ which are gauge invariant, there are two other combinations of quantities in (40.8)-(40.13) that are gauge invariant. Show that all these make six gauge invariant degrees of freedom in total (this makes sense, since four can be eliminated by fixing the gauge).
- (e) You were asked in the last homework problem set to show that for metric (40.1) the Riemann tensor is

$$R_{\mu\nu\rho\sigma} = \frac{1}{2} \left(\partial_\rho \partial_\nu h_{\mu\sigma} + \partial_\sigma \partial_\mu h_{\nu\rho} - \partial_\rho \partial_\mu h_{\nu\sigma} - \partial_\sigma \partial_\nu h_{\mu\rho} \right) . \quad (40.15)$$

Show that the Ricci tensor is

$$R_{\mu\nu} = \frac{1}{2} \left(\partial_\mu \partial^\alpha h_{\alpha\nu} + \partial_\nu \partial^\alpha h_{\alpha\mu} - \square h_{\mu\nu} - \partial_\mu \partial_\nu h \right), \quad (40.16)$$

where indices are defined to be raised by the Minkowski metric $\partial^\alpha = \eta^{\alpha\beta} \partial_\beta$, d'Alembertian is an ordinary flat space one $\square = \partial^2 = \eta^{\alpha\beta} \partial_\alpha \partial_\beta$, and the trace is $h = \eta^{\alpha\beta} h_{\alpha\beta}$.

- (f) Using the fact that the Einstein equation in vacuum reduces to $R_{\mu\nu} = 0$, write out all the equations in terms of $\phi, S, \psi, E, B_i, F_i$ and h_{ij}^{TT} and show that only the gauge invariant combinations of these quantities appear in the equations of motion.
- (g) Now we show that the only dynamical quantity in these equations is h_{ij}^{TT} .

By taking appropriate linear combinations of the components of the Einstein equation show that

$$\nabla^2 \psi = 0. \quad (40.17)$$

If we assume that the perturbation vanishes at infinity, this equation has only one solution (remember electrostatics). From here the equations and solutions for the other non-dynamical gauge invariant perturbations follow. Show that the only equation that remains is

$$\square h_{ij}^{TT} = 0. \quad (40.18)$$

Therefore, h_{ij}^{TT} represents the two physical degrees of freedom of gravity in the absence of matter.